Chapter 2. Fields Extensions

Theorem 2.5. Let E/F be a field extension, $\alpha \in E$, and $\varphi_\alpha : F[x] \to E$ the evaluation homomorphism of $F[x]$ into E such that $\varphi_\alpha(a) = a$ for all $a \in F$ and $\varphi_\alpha(x) = \alpha$. Then α is transcendental over F if and only if φ_α is a monomorphism.
Theorem 2.6. Let E/F be a field extension and $u \in E$ algebraic over F. If $m \in F[x]$ is a monic polynomial of minimal degree such that $m(u) = 0$, then (i) m is irreducible in $F[x]$; (ii) for any $f \in F[x]$, $f(u) = 0$ if and only if m divides f; (iii) m is uniquely determined by u.
Corollary 2.23. Let $F_1, F_2, \dots, F_r$ be fields, and suppose that $F_{i+1} : F_i$ is a finite extension for $i = 1, \dots, r-1$. Then

$$[F_r : F_1] = [F_r : F_{r-1}]\,[F_{r-1} : F_{r-2}] \cdots [F_2 : F_1].$$

Corollary 2.24. Let E/F be a field extension, $u \in E$ algebraic over F, and $v \in F(u)$. Then $\deg(v, F)$ divides $\deg(u, F)$.
Corollary 2.31. Let $K/E/F$ be field extensions. Then K/F is algebraic if and only if both K/E and E/F are algebraic.
Corollary 2.32. Let E/F be a field extension. Then

$$A = \{\, u \in E \mid u \text{ is algebraic over } F\}$$

is a subfield of E, and A/F is an algebraic extension.

Chapter 4. Splitting Fields

Theorem 4.5. Let f be a polynomial of degree $n \geq 1$ over a field F. Then a splitting field $E \supseteq F$ of f over F always exists, and $[E : F] \leq n!$.
Theorem 4.11. Let $\varphi : F \to F'$ be a field isomorphism. Given a monic irreducible polynomial $p \in F[x]$, let u be a root of p in some extension $E \supseteq F$ and let v be a root of p^φ in some extension $E' \supseteq F'$. Then there is a unique isomorphism $F(u) \to F'(v)$ given by

$$f(u) \;\mapsto\; f^\varphi(v) \quad (f \in F[x]),$$

that extends φ and sends u to v.
Corollary 4.12. Let p be a monic irreducible polynomial and let u, v be two roots of p in extension fields $E, E' \supseteq F$. Then $F(u) \cong F(v)$. Moreover, if $\deg(u, F) = n$, the map

$$a_0 + a_1 u + \cdots + a_{n-1} u^{n-1} \;\longmapsto\; a_0 + a_1 v + \cdots + a_{n-1} v^{n-1}$$

is an isomorphism $F(u) \to F(v)$ sending $u \mapsto v$ and fixing every element of F.
Theorem 4.13. Let $\varphi : F \to F'$ be an isomorphism of fields, and let $f \in F[x]$ be a non-constant polynomial with $f^\varphi \in F'[x]$ its image under φ. If $E \supseteq F$ is a splitting field of f and $E' \supseteq F'$ is a splitting field of f^φ, then there exists an isomorphism $E \to E'$ extending φ.
Corollary 4.14. Any two splitting fields of a polynomial $f(x) \in F[x]$ over F are isomorphic.
Theorem 4.20. A finite extension E/F is normal if and only if E is a splitting field of some polynomial in $F[x]$.
Corollary 4.21. Let E/F be a finite normal extension and let L be an intermediate field ($F \subseteq L \subseteq E$). Then any F-monomorphism $L \to E$ can be extended to an F-automorphism of E.
Corollary 4.23. Let E/F be a finite normal extension. If u and v are roots in E of the same irreducible polynomial over F, then there exists an F-automorphism θ of E such that $\theta(u) = v$.

Chapter 5. Finite Fields

Theorem 5.3. The characteristic of a unital ring R coincides with the order of the element 1 in the additive group $(R, +)$. More precisely, if $\mathrm{ord}(1) = \infty$ in $(R, +)$ then $\mathrm{char}(R) = 0$, while if $\mathrm{ord}(1) = n$ then $\mathrm{char}(R) = n$.
Theorem 5.4. The characteristic of an integral domain is either 0 or a prime number.
Corollary 5.5. The characteristic of a field is either 0 or a prime number.
Theorem 5.7. Let E/F be a finite field extension of degree n. If F has q elements, then E has q^n elements.

Corollary 5.8. If E is a finite field of characteristic p, then $|E| = p^n$ for some integer $n \geq 1$.
Theorem 5.9. Let E be a field with $|E| = p^n$, where p is prime. Then the elements of E are precisely the roots of the polynomial $x^{p^n} - x$ in $\mathbb{Z}_p[x]$, and E is the splitting field of this polynomial over $\mathbb{Z}_p$.
Theorem 5.11. The multiplicative group F^* of nonzero elements of a finite field F is cyclic.
Corollary 5.12. A finite extension E of a finite field F is a simple extension of F.
Theorem 5.14. Let F be a field and let $f, g \in F[x]$. Then: (1) $D(af) = a\,(Df)$ for all $a \in F$. (2) $D(f + g) = Df + Dg$. (3) $D(fg) = (Df)\,g + f\,(Dg)$. (4) $D(f(g(x))) = Df(g(x)) \cdot Dg(x)$.
Theorem 5.16. Let f be a polynomial in $F[x]$ and let L be a splitting field of f over F. Then the roots of f in L are all distinct if and only if f and its formal derivative Df have no nonconstant common factor.
Theorem 5.17. Let p be a prime and $n \geq 1$ an integer. Then there exists, up to isomorphism, exactly one field of order p^n.
Theorem 5.19. As an additive group, $\mathrm{GF}(p^n) \cong \mathbb{Z}_p \oplus \cdots \oplus \mathbb{Z}_p$ (n summands). As a multiplicative group, the nonzero elements $\mathrm{GF}(p^n)^*$ form a cyclic group of order $p^n - 1$, i.e. $\mathbb{Z}_{p^n-1}$.
Theorem 5.20. Let F be a field and let $f \in F[x]$ be an irreducible monic polynomial of degree $m \geq 1$. Then the field $F[x]/\langle f \rangle$ can be represented as

$$\{a_0 + a_1\alpha + \cdots + a_{m-1}\alpha^{m-1} \mid a_i \in F,\ f(\alpha) = 0\},$$

where $\alpha = x \bmod f$. Moreover, this representation of elements is unique:

$$a_0 + a_1\alpha + \cdots + a_{m-1}\alpha^{m-1} \;=\; b_0 + b_1\alpha + \cdots + b_{m-1}\alpha^{m-1}$$

if and only if $a_i = b_i$ for each i.
Corollary 5.25. $\mathrm{GF}(p^n) = \mathbb{Z}_p(u)$, where u is a primitive element of $\mathrm{GF}(p^n)$, i.e., a generator of the cyclic group $\mathrm{GF}(p^n)^*$.
Corollary 5.26. If p is a prime and $n \geq 1$, then there exists an irreducible polynomial of degree n over $\mathbb{Z}_p$.

Chapter 6. Galois Theory

Lemma 6.3. Let E/F be a field extension, $u \in E$, and $\sigma \in \mathrm{Gal}(E/F)$. Then: (i) $\sigma(f(u)) = f(\sigma(u))$ for all $f \in F[x]$. (ii) If u is a root of some $f \in F[x]$, then $\sigma(u)$ is also a root of f. (iii) If u is algebraic over F and $\sigma, \tau \in \mathrm{Gal}(F(u)/F)$, then $\sigma = \tau$ if and only if $\sigma(u) = \tau(u)$.
Theorem 6.4. Let $F(u)/F$ be a simple extension with u algebraic over F, and let $u_1 = u, u_2, \ldots, u_r$ be the distinct roots of $\mathrm{irr}(u, F)$ in $F(u)$. Then

$$\mathrm{Gal}(F(u)/F) = \{\sigma_1 = \mathrm{id}, \sigma_2, \ldots, \sigma_r\},$$

where each σ_i is the unique F-automorphism of $F(u)$ satisfying $\sigma_i(u) = u_i$. Hence if $\mathrm{irr}(u, F)$ splits in $F(u)$ and has distinct roots, then $|\mathrm{Gal}(F(u)/F)| = [F(u) : F]$.
Theorem 6.7. Let $E = F(u_1, \ldots, u_n)/F$ be a finite extension where each u_i is algebraic over F with minimal polynomial $m_i = \mathrm{irr}(u_i, F)$. If $\sigma \in \mathrm{Gal}(E/F)$, then: (i) For any $\tau \in \mathrm{Gal}(E/F)$, one has $\sigma = \tau$ if and only if $\sigma(u_i) = \tau(u_i)$ for each i. (ii) $\sigma(u_i)$ is a root of m_i for each i. (iii) σ is uniquely determined by the choice of $\sigma(u_1), \ldots, \sigma(u_n)$ in E. In particular, $\mathrm{Gal}(E/F)$ is a finite group.
Theorem 6.9. Let $G = \mathrm{Gal}(E/F)$ where E is the splitting field of a polynomial $f(x) \in F[x]$ and let X be the set of distinct roots of f in E. Then: (i) G is isomorphic to a subgroup of the symmetric group S_X. (ii) If $f(x)$ is irreducible over F, then G acts transitively on X. (iii) If $f(x)$ has no repeated root in E and G acts transitively on X, then f is irreducible over F.
Theorem 6.13. Let F be a field and f an irreducible polynomial in $F[x]$. (i) If $\mathrm{char}(F) = 0$, then f is separable over F. (ii) If $\mathrm{char}(F) = p > 0$, then f is separable over F unless it is of the form $f(x) = g(x^p)$ for some $g \in F[x]$. More precisely, $f(x) = g(x^p) = b_0 + b_1x^p + b_2x^{2p} + \cdots + b_mx^{mp}$.
Corollary 6.14. If $\mathrm{char}(F) = 0$, then F is a perfect field, and thus every algebraic extension of F is separable.
Theorem 6.15. Let F be a field of characteristic $p > 0$, and let

$$f(x) = g(x^p) = b_0 + b_1x^p + \cdots + b_mx^{mp}.$$

Then the following are equivalent: (i) f is irreducible in $F[x]$; (ii) g is irreducible in $F[x]$ and not all the coefficients b_i are p-th powers in F.

Theorem 6.16. Every finite field is perfect.
Theorem 6.17. Let $E \supseteq K \supseteq F$ be fields. If E/F is a finite separable extension, then E/K is also separable.
Theorem 6.18. Let $\varphi : F \to F'$ be an isomorphism of fields and let $f \in F[x]$ be a polynomial. If $E \supseteq F$ and $E' \supseteq F'$ are splitting fields of f and f^φ respectively, then the number of field isomorphisms $\hat{\varphi} : E \to E'$ extending φ is at most $[E : F]$, with equality if f is separable over F.
Corollary 6.19. Let $E \supseteq F$ be a splitting field of a polynomial $f \in F[x]$. Then $|\mathrm{Gal}(E/F)| \leq [E : F]$, with equality if f is separable over F.
Theorem 6.25. Let E/F be a finite field extension. Then $|\mathrm{Gal}(E/F)| \leq [E : F]$.
Theorem 6.36. For a finite extension E/F with Galois group $G = \mathrm{Gal}(E/F)$, the following conditions are equivalent: (i) E/F is Galois, (ii) E/F is normal and separable.
Corollary 6.37. If $\mathrm{char}(F) = 0$, then the finite Galois extensions of F are precisely the normal extensions.
Corollary 6.38. Every finite Galois extension is simple (i.e., of the form $F(\alpha)$ for some α).
Corollary 6.39. If $E \supseteq K \supseteq F$ are fields and E/F is a finite Galois extension, then E/K is Galois.
Theorem 6.41. Let E/F be a finite Galois extension with Galois group $G = \mathrm{Gal}(E/F)$. For an intermediate field K (so $F \subseteq K \subseteq E$) let $K^* = \mathrm{Gal}(E/K)$, and for a subgroup $H \subseteq G$ let

$$H^\circ = \{x \in E \mid \sigma(x) = x \ \forall \sigma \in H\}$$

be the fixed field of H. Then: (i) $H = (H^\circ)^*$ and $K = (K^*)^\circ$. (ii) The maps $K \mapsto K^*$ and $H \mapsto H^\circ$ are mutually inverse inclusion-reversing bijections between the set of intermediate fields of E/F and the set of subgroups of G. (iii) $[E : K] = |K^*|$. (iv) $[K : F] = |G : K^*|$, the index of K^* in G. (v) E/K is Galois. (vi) K/F is Galois if and only if K^* is a normal subgroup of G, in which case $\mathrm{Gal}(K/F) \cong G/K^*$.